Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/11/2020
PI Name:	Haveman, Natasha Ph.D.		
Project Title:	Plant Microbiome - Its Influence in Plant Growth and Protection Against Pathogenesis in Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: Dr. Natasha Haveman, formerly at the University of Florida (UF), joined the NASA Kennedy Space Center (KSC) Space Biology team in 2023. NOTE: name change to Haveman née Sng (Ed., June 2021)		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	02/13/2019	**End Date:**	02/12/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/12/2022 per NSSC information (Ed., 5/25/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Foster, Jamie Ph.D. (MENTOR: University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC19K0520		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A. Project Description - Long-duration exploration in space has emphasized the importance of understanding how to grow plants under microgravity, so as to supply crews with a continuous source of fresh food as well as provide psychological benefits. Plant and animal microbiomes have been known to play a significant role in maintaining host health and development through the regulation of metabolic and immune responses. Therefore, understanding the microbial population associated with plant growth and suppressing multi-host pathogenesis in spaceflight is not only crucial for sustaining the production of edible crops in space but also vital for the safety of crew members. The objective of this ground-based investigation is to develop a refined understanding of the basal plant microbiome of salad crops grown in plant growth chambers under previously used International Space Station (ISS) growth conditions and assess its potential for plant growth enhancements and human pathogenicity. Defining the basal plant microbiome will enable us to understand how spaceflight causes alterations in plant-microbial interactions and how to utilize it for future space exploration efforts. Hence, the overall goal of this project is to determine the influence of microorganisms associated with salad crops in plant growth and protection against pathogenesis under conditions similar to the ISS by utilizing metagenomics and metatranscriptomic approaches. Our central hypothesis is that changes in gene expression within the microbial population are key to understanding whether the plant microbiome can promote the growth of these crops through growth enhancement pathways. To investigate this hypothesis the following objectives were proposed: Objective 1 – to define the metatranscriptome of three different crops grown in plant growth chambers under ISS conditions. Objective 2 – to develop an understanding of the metabolic pathways and signaling leading to plant growth enhancements and/or to plant/human pathogenesis identified from the metatranscriptomic data. Objective 3 – to develop methods for enhancing plant growth or suppressing the pathogenic activity without eradicating commensal or saprophytic microbes. B. Significance - The rationale behind this proposal is to advance the technical readiness necessary to support a sustainable source of fresh food production for extraterrestrial exploration, and this will be accomplished by establishing a strong foundation in understanding how plant-microbe interactions impact the safe and successful production of plants. At the completion of these studies, it is expected that information gathered on the basal microbiome of salad crops cultivated in growth chambers similar to environmental conditions on the ISS will identify those microbes that have the potential to result in multi-host pathogenesis. The proposed metatranscriptomic studies, together with the metagenomic studies conducted by the space food production team at Kennedy Space Center (KSC), will help elucidate the functional pathways and mechanisms by which microbes interact with their host under ISS growth conditions. These data will then allow us to make crucial and informed engineering choices on plant growth hardware systems and crop selection for future space exploration vehicles and habitats. Furthermore, this study will provide new insights into standard terrestrial plant biology as little is currently known about the microbiome of salad crops.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There exists a limited amount of knowledge about how the vast majority of plant-associated microorganisms impacts the plant's growth, health, and susceptibility to diseases. To date, the interplay between plants and microorganisms has been largely studied for various leaf pathogens, symbiotic rhizobia, and mycorrhizal fungi. However, little is known about the microbiome of salad crops. Salad crops require minimal processing before consumption and are used to investigate the baseline microbial data for developing space-related produce sanitation methods for safe consumption by the crew. So far, microbial studies performed on salad vegetables, such as spinach, lettuce, and cilantro, are mostly culture-dependent methods to determine whether human pathogens survive and persist on the edible surface of the leaves. A survey of the salad microbiome and how it plays a role in maintaining the host's health and development is lacking. Since, investigations conducted in spaceflight have revealed that microgravity does alter the plant-microbe interaction at both the genetic and physiological level with a tendency to increase the susceptibility to plant pathogenesis. It is therefore necessary that a cultivation-independent analysis of the microbial communities found on salad crops be investigated with multi-omics, predominantly metagenomics and metatranscriptomics.
	Task Progress of Year 2: In the second year, we have one conference presentation derived from this project, a submitted manuscript, and a manuscript in preparation. Manuscripts Sng NJ, Khodadad C, Dixit A, Massa G, Louyakis A, Venkateswaran K, Foster JS. Evaluating the lettuce metatranscriptome with MinION sequencing for future spaceflight food production applications. Nature Microgravity (Submitted). [Ed. note 6/18/21: article now published; see Bibliography section] Sng NJ, Khodadad C, Dixit A, Massa G, Venkateswaran K, Foster JS. Comparative Metatranscriptomics of the lettuce microbiome reveals organ-specific mutualisms. (In preparation). Oral Presentations Sng NJ, Khodadad C, Dixit A, Massa G, Louyakis A, Venkateswaran K, Foster JS. Plant-Microbial Metatranscriptomic Using Nanopore Sequencing Platform for Spaceflight Applications. 2020 American Society for Gravitational and Space Research (ASGSR), Virtual (Technical session: Plants) Research Progress for each Objective: Objective 1 – to define the metatranscriptome of three different crops grown in plant growth chambers under ISS conditions. Major Outcomes: In YR2, the major outcomes of this objective were: 1) The 28-day old lettuce metatranscriptome was established using the MinION sequencing platform. 2) Illumina NOVAseq sequencing of all 24 lettuce samples was completed. 3) High-quality RNA has been extracted from Mizuna samples and optimization for sequencing is underway. Details of work progress: To ensure high-quality metatranscriptomic analysis, a pilot study of three 28-day old lettuce samples were used to optimize a protocol for RNA extraction, sample processing, sequencing, bioinformatics pipeline, and metatranscriptomic analysis. In YR2 of the project, a thorough analysis of the pilot 28-day old lettuce leaf

Task Progress:	metatranscriptome was completed and a paper has been submitted for peer review. In parallel, the full set of 24 lettuce samples (sanitized root and leaf samples at 7, 14, 21, and 28-days old) were sequenced via the Illumina NOVAseq 6000 platform. Sequencing was performed at the University of Florida, Interdisciplinary Center for Biotechnology Research (ICBR) and has been completed. The data obtained were run through an optimized bioinformatics pipeline (completed in YR1) and preliminary data were obtained. The principal component analysis of the data shows that root and leaf samples vary in their differential gene expression profile and should be analyzed separately. A preliminary analysis of the lettuce leaf and root microbiome at various developmental time points has been established. In addition, a pilot study to optimize the sample processing method for the mizuna plants is on-going. Findings from YR1 have shown that crop-specific ribosomal probes will need to be designed for the various host organisms to improve the removal of host RNAs and facilitate the microbial RNA enrichment process. Hence, RNA from 28-day old Mizuna leaf tissues have been extracted and several different commercially available ribosomal probes were tested. These samples have been shipped to a collaborator at Cornell for sequencing and data obtained will be used to validate the most effective method to enrich for the Mizuna microbial RNA. Objective 2 – to develop an understanding of the metabolic pathways and signaling leading to plant growth enhancements and/or to plant/human pathogenesis identified from the metatranscriptomic data. Major Outcomes: In YR2, the major outcomes of this objective were: 1) the metabolic pathways of the 28-day old lettuce leaf microbiome have been established, 2) Presence of potential bacterial genera and metabolic activity that could play a role as plant growth enhancers and plant pathogens were revealed, and 3) Gene network analysis identified several genes within the fungal MAPK pathway that could be drivers of fungal pathogenesis. Details of work progress: In YR2, we have established the microbial populations found in the 28-day old lettuce leaves and the metabolic pathways that these microbes are utilizing. Based on the 16S and 18S taxonomy results, several bacterial genera identified were potential plant growth-promoting bacteria and fungal plant pathogens. A total of 106 KEGG level 3 pathways were identified in the bacterial population and 85 in the fungal population. Metatranscriptomic analyses of the bacterial and fungal community on 28-day-old lettuce leaf tissues are largely consistent with the core functional microbiome in other phyllosphere microbial systems reported in the literature. Within the bacterial heatmap, glyoxylate and dicarboxylate metabolism and oxidative phosphorylation had the highest number of read counts with >4000 counts across all three samples. The top 15 bacterial KEGG level 3 pathways ranked based on the percentage of read counts across the samples against the total read count in all 106 pathways are as follows: Glyoxylate and dicarboxylate metabolism (47.64%), Oxidative phosphorylation (40.33%), Cell cycle – Caulobacter (1.24%), Ribosome (0.91%), Purine metabolism (0.78%), Pyrimidine metabolism (0.75%), RNA polymerase (0.56%), Aminoacyl-tRNA biosynthesis (0.43%), RNA degradation (0.41%), Two-component system (0.33%), Glycolysis / Gluconeogenesis (0.32%), Cysteine and methionine metabolism (0.32%), Pyruvate metabolism (0.26%), and Plant-pathogen interaction (0.26%). These top 15 bacterial KEGG level 3 pathways make up 94.28% of the reads assigned to the 106 pathways in the heatmap. In the fungal metatranscriptome heatmap, the top 15 KEGG level 3 fungal pathways based on the percentage of read counts across the samples against the total read count in all 85 pathways are as follows: Thermogenesis (30.43%), Oxidative phosphorylation (30.25%), Endocytosis (4.42%), Spliceosome (3.37%), Ribosome (3.19%), Protein processing in the endoplasmic reticulum (3.17%), MAPK signaling pathway (3.17%), Phospholipase D signaling pathway (1.20%), MAPK signaling pathway - yeast (0.93%), Two-component system (0.74%), Glyoxylate and dicarboxylate metabolism (0.69%), Oocyte meiosis (0.67%), FoxO signaling pathway (0.63%), Rap1 signaling pathway (0.63%), and Peroxisome (0.62%). These top 15 fungal KEGG level 3 pathways make up 84.11% of the reads assigned to the 85 pathways in the heatmap. In the fungal metatranscriptome, about a quarter of all the KEGG level 3 pathways were associated with the signal transduction pathway. Genes within our metatranscriptomic dataset that were mapped to the KEGG database showed the presence of many key genes associated with the MAPK signaling pathway in yeast. These genes also overlap with fungal pathogenesis and it would be beneficial for the crew to monitor these genes over time or under stress. Objective 3 – to develop methods for enhancing plant growth or suppressing the pathogenic activity without eradicating commensal or saprophytic microbes. In YR2, the results obtained from the 28-day old lettuce leaf microbial metatranscriptome analysis have led to a potential strategy for the suppression of fungal plant pathogens. The taxonomy diversity data revealed the presence of several potential plant growth-promoting bacteria (such as Paenibacillus) and the presence of a couple of fungal plant pathogens (such as Fusarium). Several studies have shown that Paenibacillus produces an antifungal compound that is active against fusarium. So perhaps, one way to combat the fusarium presence is to enhance the presence of paenibacillus in the population. We are investigating different molecular methods to test this hypothesis.
Bibliography Type:	Description: (Last Updated: 08/03/2023)
Abstracts for Journals and Proceedings	Sng NJ, Khodadad C, Dixit A, Massa G, Louyakis A, Venkateswaran K, Foster JS. "Plant-Microbial Metatranscriptomic Using Nanopore Sequencing Platform for Spaceflight Applications." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. (Technical session: Plants). Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Articles in Peer-reviewed Journals	Haveman NJ, Khodadad CL, Dixit AR, Louyakis AS, Massa GD, Venkateswaran K, Foster JS. "Evaluating the lettuce metatranscriptome with MinION sequencing for future spaceflight food production applications." npg Microgravity. 2021 Jun;7:22. https://doi.org/10.1038/s41526-021-00151-x ; PMID: 34140518 , Jun-2021